

Homework 1

due 18.10.2023, 14:00

How to Submit and General Rules

- Solve the assignment: write the code(s) and submission script(s), submit the necessary jobs and gather the outputs. Create graphs if necessary.
- Download the $\text{\LaTeX}$ template. Replace the included codes, skripts and graphs by yours. Write answers to questions.
- Submit your solution via moodle. The submission deadline is strict and the system will not accept late submissions. Any member of the group can make the submission.
- Submissions must be original work. Plagiarisms (even partial ones) will be punished and can lead to exclusion from the course.
- There will be 10 homeworks with 10 points each. You will need to collect at least 50 points.

Exercise 1.1 - Trivial Parallelization (4 points)

Use the “trivial parallelization” example from the lecture to distribute the same problem onto 1,2,3,... 10 tasks. Plot the timing results in the following way: On the x-axis the number of tasks, on the y-axis the “speedup”. The speedup when using n tasks is the time that 1 task needed divided by the time that n tasks need. E.g. if 1 task takes 1 second and 3 tasks need 0.5 seconds, the speedup is 2.

Interpret what you see.

(please turn the page)

Exercise 1.2 - Stromboli Cluster (6 points)

Log into stromboli and gather information on the frontend.

- What operating system does it run?
- How many cores does it have?
- What type of processors are these?
- How many cache levels do they have and how big are they?
- How much memory does the frontend have?

Next write an MPI program that will never end by itself (infinite loop). Submit it to the queuing system. Find out on which node it runs. You can `ssh` into nodes on which you have a job running, e.g.

```
ssh node23
```

Do this and gather the same information as before for a node of the “compute2011” partition. At last, cancel the job from the queuing system, using `scancel <job-id>`.